

Pranjal Arya

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Skills

- **Languages:** C, ARM Assembly, Bash scripting.
- **Protocols / Device drivers:** USB, SD Card, SDIO, UART, SPI, I2C, and GPIO.
- **Processor:** ARM Cortex - (M4, M33).
- **Miscellaneous:** Git, GDB, Saleae analyzer, SEGGER SystemView, FreeRTOS, TrustZone for Cortex-M, Pre-silicon and post-silicon bring-up.

Experience

Member of the team responsible for firmware, device drivers, BSP, and middleware development for a family of Infineon PSoC devices.

Infineon Technologies **Senior Software Engineer** **April 2023 - Present**

- **Firmware development & bring-up:**
 - Performed Pre-silicon and post-silicon bring-up on various ARM Cortex-M4 and M33 based MCUs.
 - Development of BSP and firmware for upcoming PSoC6 MCUs.
 - Ensured smooth integration of delivered assets to middlewares, RTOS.

Infineon Technologies **Software Engineer** **July 2020 - March 2023**

- **Device drivers:**
 - Designed and developed features in USB 2.0, SD Card, SDIO, UART, SPI, I2C, and GPIO device drivers.
 - Analyzed critical issues reported by customers and internal teams and provided fixes.
- **Middleware:**
 - Involved in USB - HID middleware development.
 - Upgraded PSoC6 hardware acceleration to support the latest MbedTLS release on public GitHub.
- **BSP and CI:** Created scripts and makefiles to automate build and basic sanity testing. Integrated this framework in CI.

Cypress Semiconductor **Intern** **January 2020 - June 2020**

- **Internship project:** Unified peripheral drivers to support all PSoC6 devices and created a common peripheral driver library.

Education

B.Tech. (Electronics & Communication Engineering) - Nirma University, Ahmedabad **CPI: 8.24** | 2016 - 2020
HSC **88.9%** | 2016
SSC **89.7%** | 2014

Projects

- **Custom Real-Time Operating System development from scratch** [📄 GitHub project link](#)
Developed a custom RTOS kernel and added support for Cortex-M architecture with the following features:
 - Round-Robin scheduler
 - Heap memory manager supporting aligned memory allocation and reporting maximum memory usage statistics
 - Spinlock and semaphore support for synchronization
 - Mailbox for communication
- **Custom bootloader** [📄 GitHub project link](#)
Developed a custom bootloader for STM32L476RG MCU with the following features:
 - Host application in PC interacts with a custom bootloader over UART.
 - Capable of maintaining version information and updating application firmware using an A/B swap mechanism.
 - Integrated a secure boot mechanism to verify the authenticity of the firmware image.
- **CPU Faults on Arm Cortex-M** [📄 GitHub project link](#)
To understand ARM processor internals, generated various Arm Cortex-M4 system exceptions using software on TM4C123GXL Tiva C series MCU. Wrote a fault handler to dump the status of the processor over debug prints.

Awards

- **Global Spot Award** Infineon Technologies, January 2023
Awarded for significant contribution to device driver library development.
- **YIP Award (Infineon Idea Management Program)** Infineon Technologies, February 2023
Awarded to a team of 2 members for internal framework improvement resulting in 40% cost reduction.